



EXTENDED-SPECTRUM BETA-LACTAMASE (ESBL) PRODUCING GRAM NEGATIVE BACTERIA AMONG CLINICAL ISOLATES IN SELECTED HOSPITALS IN TACLOBAN CITY

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ABSTRACT

Primary aim of the study is to determine the prevalence of extended spectrum beta-lactamase (ESBL) producing gram negative bacteria among clinical isolates in selected hospitals in Tacloban City. The study will serve as an eye opener to the health care providers for them to carry out better treatment and management of infectious diseases in response to threat of proliferation of multi-drug resistant pathogens. The study will not only be of help to the healthcare providers but also to the patients and the whole community as well. A total of eighteen (18) Gram (-) Isolates from different hospitals consisting of twelve (12) *E. coli* and six (6) *Klebsiella* species were utilized in the study, isolated from the following samples; urine, sputum and stool. Phenotypic screening of isolates was done in the respective clinical laboratories of the different hospitals while the genotypic screening using polymerase chain reaction (PCR) was done at the Philippine Science High School molecular laboratory. All bacterial isolates regardless whether positive or negative for phenotypic screening were screened for ESBL production through molecular testing using multiplex PCR for TEM, SHV and CTX-M genes. Among the different Isolates, six (50%) of *E. coli*, and two (33.3%) of the *Klebsiella* species were phenotypically ESBL's producer or positive, resulting to an over-all prevalence of around forty-four percent (44.4%). On

the other hand, the overall prevalence of ESBL using genotypic screening was reported at 77.7 % (14/18), with the following aggregates of genes detected: the blaTEM gene only detected in 6% (1/18); the blaCTX-M in 44% (8/18); the blaSHV in 22 % (4/18); the blaCTX-M and blaTEM in 6% (1/18), then 22 % (4/18) were negative for any of the resistance genes. From the eighteen (18) isolated bacteria, eight (8) were reported as ESBL positive in the phenotypic screening, while fourteen (14) were detected using genotypic screening (PCR), therefore, six (6) were screened negative in phenotypic detection but turned out to be positive in PCR. Using the standard formula, the sensitivity of the phenotypic screening in detecting the true ESBL's producing bacteria is at around fifty-seven percent (57.1%), which is relatively low. This means that, using the usual phenotypic screening, which is commonly used in Clinical Laboratories, will missed out potentially harmful resistant strain of Gram (-) bacteria which may affect the treatment and management of the disease; however, its specificity is at one hundred percent (100%). Based on the data obtained, a relatively high prevalence of ESBL's was observe which is an indication of proliferation of these multi-drug resistant strains, coupled with the relatively low sensitivity of phenotypic screening commonly used in most Clinical Laboratories, these may have a considerable impact in the healthcare delivery. From the conclusions drawn, the following are recommended; Standard Operating Procedure on infection control must be strictly implemented among hospitals, education of medical and nursing staffs, patients, and other stakeholders through symposium, seminar, workshops and other fora is also strongly recommended, and prudent prescription of antibiotics must be done.

Keywords: *ESBLs, PCR, Phenotypic and Genotypic screening, Sensitivity and Specificity*

INTRODUCTION

The emergence and spread of extended spectrum Beta-lactamase (ESBL) producing gram negative bacteria have increased all over the world, particularly in hospital settings. Extended Spectrum Beta-Lactamase (ESBL) producing gram negative bacteria is a multi-drug resistant and a rapidly evolving group of Beta-lactamases which share the ability to hydrolyze third-generation cephalosporins, leaving only a few reliable therapeutic options to an infection.

Extended-spectrum β -lactamase (ESBL)-producing *Escherichia coli* is an increasingly important group of community pathogens worldwide. These organisms are frequently resistant to many of the antimicrobial agents usually recommended for the treatment of infections caused by *E coli* or any other Gram-negative bacteria, such as penicillin, cephalosporins, fluoroquinolones, and trimethoprim-sulfamethoxazole. Data concerning risk factors, clinical features, and therapeutic options for such infections are scarce.

According to Gindri, I. et al. (2014). antibiotic resistance among gram negative bacilli is a rapidly expanding problem due to the organism's ability to mutate, and to acquire and transmit plasmids and other mobile genetic elements encoding resistance genes. Plasmids determine species of beta-lactamase in enteric bacteria and show resemblance to

enzymes and have the ability to hydrolyze some substrates, isoelectric point, immunological specificity and susceptibility to inhibition by plasmid-mediated beta-lactamases and differ from the metallo beta-lactamases.

Transmission of ESBLs could be from touching water or dirt that contains the bacteria, this is especially possible with water or soil that's been contaminated with human or animal fecal matter. Certain infections that can also develop resistance to antibiotics can increase the risk of getting a bacterial infection with ESBLs, such as [MRSA](#) infection. In some cases, it may spread by carrier individuals, this is called colonization which means the body is carrying the bacteria with ESBLs but isn't actively infected.

Intestinal bacteria like *Escherichia coli*, *Klebsiella pneumoniae* and related species (Enterobacteriaceae) are some of the most common causes of infectious disease, including urinary tract infection, postoperative infection, and septicemia, among others. Extended spectrum beta-lactamases (ESBL) are a group of enzymes that can be produced by species belonging to Enterobacteriaceae which have the ability to break down antibiotics belonging to the beta-lactam group (penicillin, cephalosporins) leading to infections caused by ESBL-producing Enterobacteriaceae, which often have to be treated with carbapenems, a group of beta-lactam antibiotics that has a wide antibacterial spectrum but cannot be administered orally. Mostly having the condition that associated with ESBL are urinary tract infection, diarrhea, skin infection, and pneumonia.

A similar study conducted by Kumar et al (2014), found the Occurrence of Extended Spectrum Beta-lactamases Producing Alpha Hemolytic *Escherichia coli* in Neonatal Diarrhea. *E. coli* organisms were isolated from 39 kids out of 120 samples. The assessment of isolates revealed alpha hemolytic nature of 23 isolates. The presence of enterohemorrhagic extended spectrum β -lactamases producing *Escherichia coli* in kids is a matter of concern and public health importance as it may be due to the transmission from hospital itself during the birth time or post birth admission period.

In the study conducted by Mathur et al (2002), also shows a greater percentage of *Klebsiella* species as ESBL producers. A total of 678 Gram negative bacteria were included in their study. They were isolated from various clinical samples obtained from indoor patients admitted to the All India Institute of Medical Sciences, New Delhi during March to June 2001. These isolates were screened for ESBL production by the inhibitor-based test recommended by the National Committee for Clinical Laboratory Standards (NCCLS). Out of the 678 strains tested, 458 (68%) were found to be ESBL producers. Among the bacterial species, ESBL production was most common in *Klebsiella* spp. (80%). The proportion of ESBL positive isolates was highest from intensive care units (79%), followed by Medical Oncology (75%), Medical (54%) and Surgical wards (50%).

A study conducted by Ali et al, determined the frequency of extended spectrum beta lactamase producing gram negative bacilli among clinical isolates at clinical laboratories. A total of 812 consecutive non duplicate gram-negative bacilli recovered from clinical specimens. These were isolated from various samples including urine, blood, pus

aspirate, sputum, chest tube, HVS, i/v canula/ CVP lines and catheter tips received from patients admitted in Military Hospital. Three hundred and sixty-six isolates were ESBL producers making a frequency of 45 %. *Enterobacter cloacae* was the most frequent ESBL producer. *Escherichia coli* (45%) was the most frequent organism isolated followed by *Klebsiella pneumoniae* (21%), *Pseudomonas aeruginosa* (19.2%), *Enterobacter cloacae* (4.6%) and *Acinetobacter baumannii* (4.4%).

The presence of beta-lactamases may be chromosomal encoded and universally present in a species or plasmid mediated. PBPs (Penicillin Binding Protein) are believed to produce evolved chromosomal enzymes which then show same-sequence homology. It then resulted in the selective pressure exerted by the beta-lactam producing soil organisms. The incidence of resistance to extended-spectrum β -lactamase (ESBL) becomes a major problem among Gram negative pathogens. β -lactamase production among the members of the Enterobacteriaceae family is the most common mechanism responsible for resistance to β -lactam antibiotics. Usually, their homologous sequences of amino acids divide β -lactamases into four classes (A-D). The plasmid-mediated class A β -lactamases most frequently reported are derived from TEM (called Temoniera) and SHV (called sulfhydryl variables). Datta and Kontomichalou reported for the first time in 1965 that TEM-1 (classical β -lactamase type TEM) indicates high resistance to penicillin (ampicillin and piperacillin) and early cephalosporin (cephalosporin). It was later suggested that TEM-1 (classical TEM-species β -lactamase) gives little resistance to oxyiminocephalosporins, monobactams and carbapenems.

ESBLs derived from TEM-1 appeared in the early 1980s with the introduction of β -lactam antibiotics into the medicine field. In most cases, ESBLs are plasmid-mediated, and most of them are mutant forms of TEM-1 enzymes that have replaced one or more amino acids. The active site is surrounded by these amino acids. (Paterson DL, Ko WC, Von Gottberg A, Casellas JM, Mulazimoglu L, Kaugman KP, *et al*).

In study conducted by Turner *et al* (2009), found out that Plasmid-borne *bla*_{SHV} genes in *Klebsiella pneumoniae* are associated with strong promoters. Consistent with this study, GenBank searches reveal that the large SHV transposon or modified derivatives always carry the mutated *bla*_{SHV} promoter sequence, and this has been detected on plasmids harbored by *K. pneumoniae*, *Yersinia pestis*, *Enterobacter cloacae* and *Salmonella enterica* serovar Typhimurium^{8,9} (GenBank accession numbers AY532647 and AJ245670). The study concluded that both the large and small SHV transposons possess strong *bla*_{SHV} promoters, albeit generated by different mechanisms. This suggests that SHV ESBLs are plasmid associated in *K. pneumoniae* because an ESBL-conferring mutation in the poorly expressed chromosomal gene will not confer a phenotype. The detection of the large SHV transposon in ESBL-negative isolates in this study, and also in an ESBL-negative *Y. pestis*,⁹ indicates that there has been dissemination of highly expressed, plasmid-borne non-ESBL-encoding *bla*_{SHV} genes prior to the evolution of SHV ESBLs. It may be that the mutant promoter first appeared on the chromosome and was subsequently mobilized onto a plasmid, or that there have been parallel instances of the evolution of that promoter sequence, or that the gene has integrated back into the chromosome.

One of the most widely prescribed antibiotics worldwide are the Beta-lactam antibiotics. These include the penicillins, cephalosporins, carbapenems and monobactams. These group of antibiotics acts on the cell wall of a bacterial cell specifically the cell wall synthesizing enzymes, also called the penicillin-binding proteins that catalyzes the peptidoglycan wall of the bacterium, which results in the weakening of the cell wall structure, leading to cell lysis. However, introduction of this antibacterial agent was immediately accompanied by the emergence of drug resistant organisms which adapted different mechanisms to counter the effect of the antibiotics, such as Gram-negative bacteria produce β -lactamases enzymes which are their major defense mechanism against β -lactam antibiotics.

The introduction of third-generation cephalosporins into clinical practice in the early 1980s was considered as a major breakthrough in the fight against β -lactamase-mediated bacterial resistance to antibiotics. Resistance to this extended spectrum cephalosporins emerged quickly and the first report of plasmid-encoded β -lactamase capable of hydrolyzing the extended-spectrum cephalosporins was published in 1983. Mutations occurred within the structural genes that encodes the β -lactamase enzyme giving rise to derivatives that possessed an extended substrate profile compared with that of the parental enzymes. Thus, these new enzymes were coined Extended- spectrum beta-lactamase (ESBLs) to reflect the fact that they are derivatives of older enzymes and has a new capability to hydrolyze a broader spectrum of β -lactam drugs.

The spread of ESBL-producing bacteria has expanded rapidly throughout the world, indicating the need for continuous monitoring systems and effective measures to control infections. ESBL producers have also increasingly limited therapeutic options against infections. Health care interactions including the use of antibiotics, particularly oxyiminocephalosporins, and hospital transfers are among the well-defined risk factors for the acquisition of ESBL-producing bacteria. (Kiratisin P. et al).

With the aforementioned conditions, the researchers want to evaluate the prevalence of extended spectrum beta-lactamase (ESBL) producing gram negative bacteria among clinical isolates in selected hospitals in Tacloban City. The study will act as an eye opener to the health care providers and facilitate them to carry out better treatment strategies. The study will be of help not only to the hospital care providers and patients but to the whole community as well.

Research Objectives

Detection of extended- spectrum beta-lactamase producing (ESBL) Gram-negative bacteria among clinical isolates in selected hospitals in Tacloban City.

Specifically, the study aimed:

1. To determine extended spectrum beta lactamase producing Gram-negative bacterial Isolates from selected hospitals in Tacloban City through genotypic characterization.

2. To compare the prevalence of extended spectrum beta lactamase producing bacterial isolates received from the Clinical Laboratory of the selected hospital based on the following methods
 - 2.1. Phenotypic Screening
 - 2.2. Genotypic Screening
3. To determine the Sensitivity and Specificity of phenotypic characterization of ESBL's compared to the Genetic characterization.

METHODOLOGY

Research Design

Cross-sectional research design was utilized in the study which can provide a "snapshot" of the frequency and characteristics of the subject of interest at a particular point in time.

Clinical isolates

All the identified bacterial isolates in the months of March and April, 2019 that were phenotypically screened for ESBL by the selected hospital were sub-cultured into a broth and incubated for 24 hours. After incubation, a stock culture was made by inoculating again the bacteria into another freshly made broth and incubated for 24 hours. After incubation, the broths were added with glycerol and preserved in 1-6 C temperature. This is to maintain the viability of the isolates until confirmatory test was done.

All the strains that were phenotypically screened positive and negative for ESBL production were subjected to confirmation through molecular testing. Multiplex PCR for TEM, SHV and CTX-M genes was used to detect ESBL genes in bacteria.

DNA extraction

Before PCR was made, the stock cultures were inoculated into a quadrant plate to ensure the viability of the bacterial isolates. After 24 hours incubation, a single colony from the plate was inoculated to a broth and incubated for 24 hours. This served as our overnight bacterial broth culture for DNA extraction by boiling method. A volume of 200 uL of the 24 hours incubated cultures was used and the DNA was eluted in a final volume of 50 uL.

M-PCR Amplification

The M-PCR amplification was performed according to the method of Monstein et al. (2007). The primer-pair sequences used in the M-PCR assay, melting temperatures,

expected PCR and amplicon sizes (Table 1). All the primers were manufactured by Inqaba Biotechnical Industries (Pty) Ltd (Prefactoria, South Africa).

The M-PCR assay was performed using the Qiagen Multiplex PCR kit 1000 (Qiagen, Hilden, Germany) for the simultaneous detection of the SHV (745 bp), CTX-M (593 bp) and TEM (445 bp) genes. Three sets of primers specific for CTX-M, SHV and TEM genes were used, (Table 1). The 25- μ L Final reaction mixture consisted of 12.5 μ L, 2 X Qiagen Multiplex PCR Master mix (HotStart Taq DNA polymerase, Multiplex PCR buffer and dNTP Mix), Q-solution 5 x and nuclease-free water (Qiagen) and a volume of 2 μ L of the prepared DNA template. The reaction tubes were placed in a PX2 thermal cycler with the cycling conditions as follows: denaturation at 94°C for 10 min, 30 annealing at 60°C for 1 min, extension at 72°C for 1 min and final extension step at 72°C for 7 min. A 16s DNA was used as a control.

Table 1.

Nucleotide sequences of the primers used for the detection of the *blaSHV**, *blaTEM*, *blaCTX-M* genes

Primer name	Sequence direction	Melting temperature	Amplicon size (bp)	References
Bla-SHV.SE	5'-ATGCGTTATATTCGCCTGTG-3'	45	747	Monstein et al. (2007)
bla-SHV.AS	5'-TGCTTTGTTCGGGCCAA-3'	45	747	Monstein et al. (2007)
TEM-164.SE	5'-TCGCCGCATACAC-TATTCTCAGAATG-3'	53	445	Monstein et al. (2007)
TEM-165.AS	5'-ACGCTCACCGGCTCCAGATTTAT-3'	52	445	Monstein et al. (2007)
CTX-M-U1	5'-ATGTGCAGYACCAAGTAARGT-KATGGC-3'	54	593	Monstein et al. (2007)
CTX-M-U2	5'-ATGTGCAGYACCAAGTAARGT-KATGGC-3'	58	593	Monstein et al. (2007)
MA1 forward	5'-TGGGTRAARTARGTSAC-CAGAAACAGCG-3'	55	554	Pitout et al. (2005)
MA2 reverse	5'-SCSATGTGCAGYACCA-3'	55	554	Pitout et al. (2005)
	5'-CCGCRATATGRTTGGTGGTG-3'			

Detection of amplification products

The amplicons (products) of the PCR reactions were visualized using an Ultra Violet light box (UVP products), following electrophoresis on a 1.5% (m/v) agarose gel, which contained 11.5 μ L of ethidium bromide (0.5 mg mL⁻¹) (Promega, Madison). A molecular weight (50-bp ladder) marker (Promega) was used as a reference.

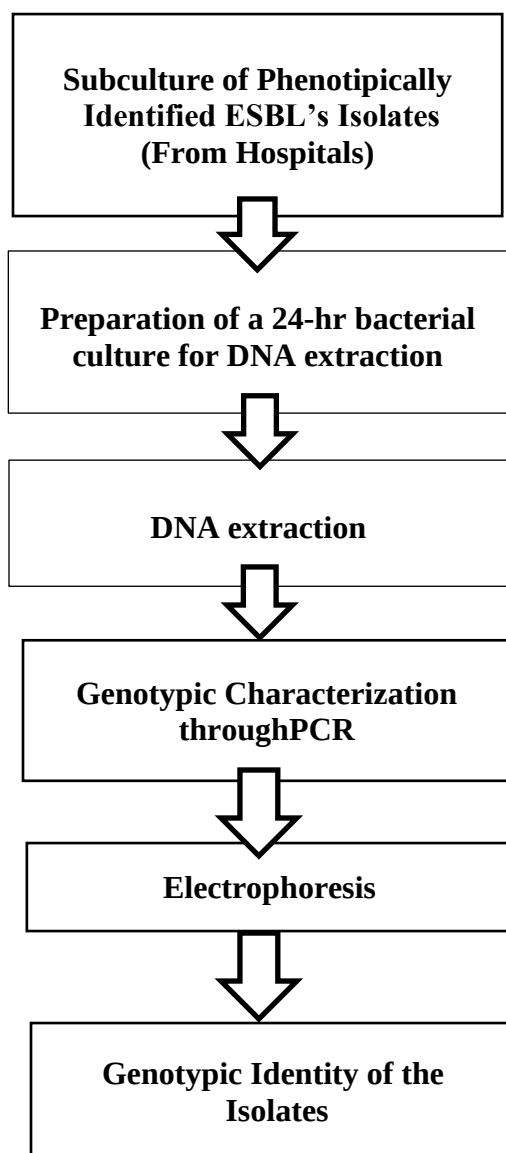


Figure 1. Schematic Diagram of Genotypic Identification of Isolate

RESULTS AND DISCUSSION

A total of 18 bacterial isolates were yielded, with the following breakdown; nine (9) from urine sample (50 %), five (5) from sputum (27.8 %), and four (4) from stool (22.2 %) as shown in table 1.

Table 2
Frequency Distribution as to the source of Specimens

Specimen	Frequency	Percentage (%)
Urine	9	50
Sputum	5	27.8
Stool	4	22.2
Total	18	100

Table 3
Frequency Distribution of Gram (-) Clinical Isolates

Isolates	Frequency	Percentage %
E.coli	12	66.7
Klebsiella	6	33.3
TOTAL	18	100

Among the isolates, majority are Escherichia coli (66.7 %) , and around thirty three percent are Klebsiella species(33.3%).

Table 4
Phenotypic Characterization of ESBL

Sample Isolate	Urine		Sputum		Stool		Total
	Fre- quency	(%)	fre- quency	(%)	fre- quency	(%)	fl(%)
E. coli	3 (8)	38	0	0	3 (4)	75	6 (50)
Klebsiella	1 (4)	25	1 (2)	50	0	0	2 (33.3)
Total	4(12)	33	1(2)	50	3(4)	75	8 (44.4)

Presented in table 4 is the frequency distribution of Gram (-) isolates based on the phenotypic ESBL characteristics in relation to the source of specimen. Six (50%) and two (33.3%) of the E. Coli and Klebsiella species isolated were phenotypically ESBL's producer or positive respectively, resulting to an over-all prevalence of around forty four percent (44.4%).

Table 5
Genotypic Characterization (PCR)

Sample Isolates	Urine		Sputum		Stool		Total
	Fre- quency	(%)	fre- quency	(%)	fre- quency	(%)	fl(%)
E. coli	5 (7)	71.4	0	0	4 (4)	80	9 (75)
Klebsiella	4 (4)	100	1 (2)	50	0	0	5 (83.3)
Total	9 (11)	81.8	1(2)	50	4(4)	100	14 (77.7)

Presented in table 5 is the genotypic characterization of which all of the 18 bacterial isolates were subjected to PCR regardless of their phenotypic results. The overall prevalence of these ESBL genes in this study was 77.7 % (14/18).

The M-PCR assay identified the results for each of the resistance genes as follows: the blaTEM gene only detected in 6% (1/18); the blaCTX-M in 44% (8/18); the blaSHV in 22 % (4/18); the blaCTX-M and blaTEM in 6% (1/18), then 22 % (4/18) were negative for any of the resistance genes (table 6).

From the isolated bacteria, eight (8) were reported as ESBL positive in phenotypic screening (table 4), however 14 were detected as ESBL positive in the genotypic characterization (table 5), therefore, six (6) bacteria that were screened negative in phenotypic detection but turned out to be positive in PCR. Three (3) out of these six showed positive for blaSHV gene which correlates with the previous studies conducted by Turner et al., suggesting that SHV ESBLs are plasmid associated wherein ESBL-conferring mutation in the poorly expressed chromosomal gene will not confer a phenotype (table 6).

Table 6
Comparison between Phenotypic and Genotypic (PCR), Characteristics of Gram (-) Isolates

Isolate	Phenotypic screening	Genotypic Screening (PCR)
E. coli 390	Negative	Negative
K. pneumoniae 398-U	Negative	Positive for blaSHV
	Positive	Positive for blaCTX-M
E. coli 333	Positive	Positive for blaCTX-M and blaTEM
K. pneumoniae 303-U	Positive	Positive for bla CTX-M
	Negative	Positive for blaSHV
E. coli 406	Positive	Positive for blaSHV
K. pneumoniae 387	Positive	Positive for blaSHV
E. coli 336	Negative	Positive for blaCTX-M
K. pneumoniae 406	Negative	Positive for blaTEM
E. coli 396	Negative	Negative
E. coli 437	Negative	Positive for blaSHV
K. pneumoniae 323-U	Negative	Negative
	Negative	Negative
E. coli 395	Negative	Positive blaCTX-M
K. pneumoniae 4219	Positive	Positive blaCTX-M
E. coli – U	Positive	Positive blaCTX-M
E. coli-S-1	Positive	Positive blaCTX-M
E. coli-S-2		Positive blaCTX-M
E. coli- S-3		
E. oli – S-4		

From the data presented in tables 4 and 5, the sensitivity and specificity of phenotypic screening compared to the genotypic screening were determined using the standard

formula for specificity and sensitivity. Based on the mathematical computations, the sensitivity of the phenotypic screening in detecting the true ESBL's producing bacteria is at around fifty seven percent (57.1%), which is relatively low. This means that, using the usual phenotypic screening, which is commonly used in Clinical Laboratories, will miss out potentially harmful resistant strain of Gram (-) bacteria which may affect the treatment and management of the disease, which eventually may cost the life of the patient. However, its specificity is at one hundred percent (100%), this means that the phenotypic screening in this particular study was able to pinpoint all the true negative.

DISCUSSION

ESBLs in Gram negative bacteria have emerged as a major problem in hospitalized as well as community-based patients. Nosocomial infections with Gram-negative bacilli are not uncommon in the local setting and can be perceived as a growing threat to public health. Since Gram-negative bacilli are the most frequently isolated organisms from clinical specimens and considering the high prevalence of ESBL production in this group there is a great need for laboratory testing methods for correctly detecting these enzymes for proper management.

The prevalence of ESBL among clinical isolates varies greatly worldwide and in geographic areas and is rapidly changing overtime. It may vary from country to country or from place to place. In our study, the incidence of ESBL was found to be 77.7% in a period of two months. In a study conducted by Siddiqui, et al in 2014 showed 61% of isolates that are found to be ESBL producer. Similarly, a study conducted by Umadevi S, et al. in India showed 69% isolates as ESBL producers.

However, studies conducted by Ali et al (45 %), Rodrigues et al (6.9%) and Menon et al (20 %), had results lower than ours. This wide variation could be explained by the high amount of geographical variation shown by ESBL producers.

ESBLs are most commonly found in *Klebsiella* spp. and *E. coli*, but they have also been detected in other members of the Enterobacteriaceae family. *Klebsiella* is the genus which frequently harbors ESBL as was seen in our study. In our study *Klebsiella pneumoniae* 83% (5/6) was the most prevalent ESBL producer as compared to *E. coli*. This result was concordant with studies conducted by Moland, et al. (11.3% *K. pneumoniae* while 2.3% *E. coli*) and Mathur et al (68%).

High percentage of ESBL producing *Klebsiella pneumoniae* in our study may be due to the selective pressure imposed by extensive use of antimicrobials. Patient to patient transfer of ESBL producing strains of *Klebsiella* spp. could also be another reason as has been shown by Jain et al in their study.

In the researchers' study, 6 isolates were found to be positive in the confirmatory test but negative in the phenotypic test. Such discrepancy between screening and confirmation test results has also been observed in other studies. This discrepancy in test results aside from the previously mentioned genes), could also be attributed to some other

additional resistance mechanisms, which include AmpC enzymes, porin changes and TEM and SHV β -lactamases with decreased affinities for β -lactamase inhibitors that can mask clavulanic acid inhibition giving false negative test results.

Conclusions

Among the eighteen (18) Gram (-) clinical isolates from the different hospitals, forty four percent were phenotypically identified to be Extended-Spectrum Beta-Lactamase producers while using genotypic screening using Polymerase Chain Reaction (PCR), around seventy seven percent were detected.

A total of six (6) from the eighteen isolates screened were failed to be detected using phenotypic screening.

Based on the data obtained, the sensitivity of phenotypic screening is relatively low at around fifty-seven percent which may have an impact in the treatment and management of the disease, while its Specificity is at its best at 100%.

Recommendations

Bacterial strains resistant to most classes of antibiotics will continue to emerge unless the inappropriate use of these drugs is curtailed. Clinicians should consider ESBL production as a possibility in case of treatment failure with Beta lactam antimicrobials, thus, prudent prescription of antibiotics must be done.

Clinical Laboratories must consider as their prime responsibility to detect and report accurately ESBLs producers for both screening and confirmatory tests.

To prevent the spread of ESBL producing organism, Standard Operating Procedure on infection control must be strictly implemented among hospitals. Development and implementation of hospital antibiotic prescribing guide should also be done.

Education of medical and nursing staffs, patients, and other stakeholders through symposium, seminar, workshops and other fora is also strongly recommended, since bacterial pathogens are constantly evolving, training must be an ongoing process.

For future researchers, the researchers of this study recommend to obtain more samples from different hospitals both public and private.

Compliance with Ethical Standards

The respondents voluntarily participated in the research study conducted by the researchers from **Doña Remedios Trinidad Romualdez Educational Foundation and Eastern Visayas State University**. The respondents understand that the research is designed to gather the data for the study. They were allowed to ask questions about the

project and their participation. Moreover, the use of the data in research, publication, sharing and archiving has been explained thoroughly. They further agreed that taking part in the project would include being surveyed, interviewed and recorded (Audio or Video) provided that their identity is kept confidential and they have the freedom to withdraw anytime. After the data were transcribed and confirmed by the respondents, the recordings were destroyed.

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